Test Project — Masonry, Plaster & Finishes BOQ Report

1. Executive Summary & Code Compliance

Item / Parameter	Value	Standard Reference
Building Footprint	12.00 m × 9.00 m	Architectural Layout
Total Stories / Height	2 Stories / 6.0 m	NBC 2016 Part 3
Total Clay Bricks	17,822 Nos	IS 1077:1992 (Modular 20×20×10 cm)
Total Mortar Cement	103 Bags (5,124 kg)	IS 269:2015 (1:4 Mix Proportion)
Total Mortar Sand	22.77 Tonnes	IS 383:2016 Zone II
Total Plaster Surface Area	712.8 m²	IS 1200 Part 12
Total Plaster Cement	58 Bags (2,852 kg)	IS 1661:1972 (12mm int, 15mm ext)
Total Plaster Sand	16.36 Tonnes	IS 383:2016 Zone III
Estimated Wall & Finishes Cost	INR 641,520.00	CPWD DSR 2023

2. Masonry Brickwork Schedule & Material Takeoff (IS 1077, IS 2250)

Bricks modular 200×200×100 mm (250 Nos/m³). Mortar 1:4 mix proportion, dry factor 1.33, W:C = 0.50.

Wall ID	Span / Cols	Len (m)	Ht (m)	Thk	Gross (m³)	Bricks (Nos)	Brick Vol	Wet Mor	Dry Mor	Cem Bags	Cem (kg)	Sand (T)	Water (L)
W_H1	C1_L0→C4_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H2	C4_L0→C7_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H3	C7_L0→C10_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H4	C2_L0→C5_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H5	C5_L0→C8_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H6	C8_L0→C11_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H7	C3_L0→C6_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H8	C6_L0→C9_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_H9	C9_L0→C12_L0	4.00	2.48	200mm	3.96	990	3.22	0.743	0.989	6	284.8	1.27	142
W_V10	C1_L0→C2_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V11	C2_L0→C3_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V12	C4_L0→C5_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V13	C5_L0→C6_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V14	C7_L0→C8_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V15	C8_L0→C9_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V16	C10_L0→C11_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
W_V17	C11_L0→C12_L0	4.50	2.48	200mm	4.46	1,114	3.62	0.836	1.111	7	320.1	1.42	160
TOTAL	All Walls	-	-	-	71.28	17,822	57.90	13.38	17.79	103	5,124	22.77	2,562

3. Plastering Schedule & Quantities (IS 1661, IS 2402, IS 1200)

Ext: 15mm 1:4 render; Int: 12mm 1:6 plaster; Ceiling: 6mm 1:3 soffit plaster per IS 1661 Cl. 12.3. Dry factor 1.33.

Surface ID	Location & Mix Spec	Area (m²)	Thk (mm)	Mix	Wet Vol (m³)	Dry Vol (m³)	Cem (Bags)	Cem (kg)	Sand (T)	Water (L)
W_H1_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H1_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H2_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H2_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H3_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H3_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H4	Int Part (2-Face 12mm 1:6)	39.6	12	1:6	0.475	0.632	3	130.0	0.87	65
W_H5	Int Part (2-Face 12mm 1:6)	39.6	12	1:6	0.475	0.632	3	130.0	0.87	65
W_H6	Int Part (2-Face 12mm 1:6)	39.6	12	1:6	0.475	0.632	3	130.0	0.87	65
W_H7_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H7_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H8_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H8_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_H9_EXT	Ext Render (15mm 1:4)	19.8	15	1:4	0.297	0.395	3	113.8	0.51	57
W_H9_INT	Int Wall (12mm 1:6)	19.8	12	1:6	0.238	0.316	2	65.0	0.43	33
W_V10_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V10_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
W_V11_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V11_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
W_V12	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V13	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V14	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V15	Int Part (2-Face 12mm 1:6)	44.6	12	1:6	0.535	0.711	3	146.3	0.98	73
W_V16_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V16_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
W_V17_EXT	Ext Render (15mm 1:4)	22.3	15	1:4	0.334	0.444	3	128.0	0.57	64
W_V17_INT	Int Wall (12mm 1:6)	22.3	12	1:6	0.267	0.356	2	73.1	0.49	37
Ceiling Soffit	Slab Bottom (6mm 1:3)	216.0	6	1:3	1.296	1.724	13	620.5	2.07	310
TOTAL	All Plaster Surfaces	928.8	-	-	10.473	13.929	70	3,473	18.43	1,736

4. Painting & Surface Finishes Schedule (NBC 2016 SP7)

1 coat primer @ 8 m²/L, 2 coats polymer putty @ 1.5 kg/m², 2 coats emulsion (interior 12 m²/L, exterior 10 m²/L).

Surface ID	Location & Specification	Finish Area (m²)	Primer 1-Ct (L)	Putty 2-Ct (kg)	Int Emulsion 2-Ct (L)	Ext Weather 2-Ct (L)
W_H1_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H1_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H2_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H2_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H3_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H3_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H4	Internal Partition Emulsion	39.6	5.0	59.4	6.6	—

Surface ID	Location & Specification	Finish Area (m²)	Primer 1-Ct (L)	Putty 2-Ct (kg)	Int Emulsion 2-Ct (L)	Ext Weather 2-Ct (L)
W_H5	Internal Partition Emulsion	39.6	5.0	59.4	6.6	—
W_H6	Internal Partition Emulsion	39.6	5.0	59.4	6.6	—
W_H7_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H7_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H8_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H8_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_H9_EXT	External Weather Guard	19.8	2.5	—	—	4.0
W_H9_INT	Internal Acrylic Emulsion	19.8	2.5	29.7	3.3	—
W_V10_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V10_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
W_V11_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V11_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
W_V12	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—
W_V13	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—
W_V14	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—
W_V15	Internal Partition Emulsion	44.6	5.6	66.8	7.4	—
W_V16_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V16_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
W_V17_EXT	External Weather Guard	22.3	2.8	—	—	4.5
W_V17_INT	Internal Acrylic Emulsion	22.3	2.8	33.4	3.7	—
Ceiling Soffit	Slab Underside (White Emulsion)	216.0	27.0	324.0	36.0	—
TOTAL	All Surfaces (Walls + Ceilings)	928.8	116.1	1,081.4	120.1	41.6

5. Consolidated Material Takeoff Summary (All Stories)

Item / Material Description	Specification / Mix Proportion	Building Total	Unit	IS Standard
Total Wall Gross Volume	L × t × Hv (clear story height)	71.28	m³	IS 1905:1987
Clay Modular Building Bricks	Standard modular 20×20×10 cm	17,822	Nos	IS 1077:1992
Net Solid Brickwork Volume	Actual brick body volume (excl. mortar)	57.90	m³	IS 1077:1992
Wet Brickwork Mortar Volume	Gross Wall Volume – Solid Brick Vol	13.38	m³	IS 2250:1981
Dry Brickwork Mortar Volume	Wet Volume × 1.33 dry factor	17.79	m³	IS 2250:1981
Brickwork Cement	1:4 Mix Proportion (OPC / PPC)	103 (5,124 kg)	Bags (50kg)	IS 269:2015
Brickwork Sand	Coarse sand, dry density 1600 kg/m³	22.77	Tonnes	IS 383 Zone II
Brickwork Water	Water:Cement ratio = 0.50	2,562	Litres	IS 456 Cl 5.4
Wall Plastering Surface Area	Internal (12mm 1:6) + External (15mm 1:4)	712.8	m²	IS 1200 Pt 12
Ceiling Soffit Plaster Area	Soffit bottom of slabs (6mm 1:3 mix)	216.0	m²	IS 1661:1972
Total Plaster Surface Area	Combined wall + ceiling surface	928.8	m²	IS 1200 Pt 12
Plastering Cement	Combined wall & ceiling plaster	70 (3,473 kg)	Bags (50kg)	IS 1661 / IS 2402
Plastering Sand	Clean plaster sand (Zone III/IV)	18.43	Tonnes	IS 383:2016
Plastering Water	Water:Cement ratio = 0.50	1,736	Litres	IS 456 Cl 5.4
Surface Primer	Water-thinnable primer @ 8 m²/L (1 coat)	116.1	Litres	NBC 2016 SP7

Item / Material Description	Specification / Mix Proportion	Building Total	Unit	IS Standard
Polymer Wall Putty	2 coats (1–2 mm) @ 1.5 kg/m²	1,838.7	kg	NBC 2016 SP7
Interior Acrylic Emulsion	2 coats (30–40 microns) @ 12 m²/L	120.2	Litres	NBC 2016 SP7
Exterior Weather Emulsion	2 coats (40–60 microns) @ 10 m²/L	41.6	Litres	NBC 2016 SP7
GRAND TOTAL CEMENT (Mortar + Plaster)	All masonry + plastering works combined	173 bags (8,596 kg)	Bags	IS 269:2015
GRAND TOTAL SAND (Mortar + Plaster)	All masonry + plastering sand combined	41.20	Tonnes	IS 383:2016
GRAND TOTAL WATER (Mortar + Plaster)	All masonry + plastering water combined	4,298	Litres	IS 456 Cl 5.4

6. Trade-wise Cost Estimate (CPWD DSR)

Trade / Material Component	Quantity	Rate (INR)	Subtotal (INR)
Clay Modular Bricks (IS 1077)	17,822 Nos	■9 / brick	■160,398.00
Brickwork Mortar Cement	103 Bags	■380 / bag	■39,140.00
Brickwork Mortar Sand	22.77 Tonnes	■1,500 / Tonne	■34,157.78
Plastering Work (12/15/6mm)	928.8 m²	■280 / m²	■260,064.00
Surface Finishes (Primer + Putty + Paint)	928.8 m²	■120 / m²	■111,456.00
TOTAL MASONRY, PLASTER & FINISHES			■605,215.78

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